

Drug Abuse in Schizophrenic Patients: Clinical Correlates and Reasons for Use

Lisa Dixon, M.D., Gretchen Haas, Ph.D., Peter J. Weiden, M.D.,
John Sweeney, Ph.D., and Allen J. Frances, M.D.

Objective: This study aimed to 1) determine substance abuse prevalence and preference in a diverse sample of schizophrenic, schizoaffective, and schizophreniform inpatients, 2) compare drug-abusing and non-drug-abusing patients on demographic and clinical variables during the acute and stabilization phases of their hospital course, and 3) obtain data from patients on reasons for drug abuse and on acute state-related changes during periods of intoxication. **Method:** Eighty-three psychotic inpatients consecutively admitted to a New York City teaching hospital were evaluated. Sixty-eight had schizophrenia, 12 had schizoaffective disorder, and three had schizophreniform disorder diagnosed according to the Structured Clinical Interview for DSM-III-R. Each patient received ratings on the Brief Psychiatric Rating Scale, the Global Assessment Scale, and the Scale for the Assessment of Negative Symptoms at admission and at discharge, an evaluation of premorbid adjustment, and an extensive interview on drug and alcohol use. **Results:** Forty (48%) of the patients received diagnoses of drug or alcohol abuse or dependence. The drug-abusing patients primarily used cannabis (N=26), alcohol (N=21), and cocaine (N=14) and reported that they abused drugs to get "high," to relieve depression, and to relax. They had significantly fewer positive and negative symptoms at discharge, better sexual adjustment and worse school performance during adolescence, and more family histories of drug abuse than the non-drug-abusing patients. **Conclusions:** Schizophrenic patients who abuse drugs may represent a subgroup of patients with better prognoses and less severe clinical characteristics of schizophrenia, but their drug abuse may adversely affect global outcome.
(Am J Psychiatry 1991; 148:224-230)

A substantial portion (15%–60%) of schizophrenic patients abuse psychoactive drugs (1–4). While clinicians have become increasingly aware of the scope of the problem, little is known about the clinical epidemiology and causes of substance abuse in schizophrenic patients. Such information may be clinically useful and perhaps offer insight into the heterogeneity and pathogenesis of schizophrenia.

Several models have been proposed to explain the frequent comorbidity of substance abuse and schizophrenia. The vulnerability model hypothesizes that

drug abuse may cause schizophrenia or increase the likelihood of its expression in an already vulnerable individual. McLellan et al. (5) found that amphetamine abusers studied over 6 years were significantly more likely than barbiturate abusers to develop psychotic disorders. Andreasson et al. (6) reported that Swedish conscripts who were high consumers of cannabis had a relative risk of schizophrenia that was six times that of nonusers. It has been found that drug-abusing schizophrenic patients are younger (1, 2, 7), have the onset of schizophrenia at earlier ages, and have better premorbid adjustment (8–11) than non-drug-abusing schizophrenic patients—findings suggesting that drug abuse may trigger the development of schizophrenia.

A second model postulates that drug-abusing schizophrenic patients may be self-medicating their symptoms. This model implies a selective matching of specific substances (and their subjective effects) with specific symptoms, a match that may have informative psychobiological origins. For example, drug abuse may be understood as an "on-the-street" parallel to

Presented at the 142nd annual meeting of the American Psychiatric Association, San Francisco, May 6–11, 1989, and at the International Congress of Research in Schizophrenia, San Diego, April 1–5, 1989. Received Jan. 30, 1990; revision received May 4, 1990; accepted Aug. 20, 1990. From the Payne Whitney Clinic, New York. Address reprint requests to Dr. Dixon, Maryland Psychiatric Research Center, P.O. Box 21247, Baltimore, MD 21228.

Supported by NIMH grants MH-43613 and MH-42969, a grant from the Scottish Rite Schizophrenia Research Fund, and a grant from the Stephen Levy Foundation.

Copyright © 1991 American Psychiatric Association.

the laboratory finding that amphetamine challenge tests improve the mood and increase the work efficiency of some schizophrenic patients (12–14). A review of retrospective studies (15) has suggested that schizophrenic patients prefer drugs which might counteract negative symptoms (e.g., cocaine, cannabis, amphetamines, and hallucinogens) rather than potentially depressant drugs, such as sedative-hypnotics, opiates, and alcohol. A third, related model postulates that drug-abusing patients self-medicate uncomfortable neuroleptic-induced extrapyramidal symptoms (3, 15, 16). Anecdotal reports support the notion that schizophrenic patients self-medicate both negative symptoms and extrapyramidal symptoms by using psychoactive drugs (16).

Our study evaluated substance abuse in consecutively admitted schizophrenic, schizoaffective, and schizophreniform disorder inpatients. We used prospectively gathered data to conduct a multidimensional comparison of drug-abusing and non-drug-abusing patients on serial clinical dimensions. To our knowledge, no prospective study of inpatients of such size and scope has been previously reported.

The study aimed to 1) determine substance abuse prevalence and preference in a diverse sample of these inpatients, 2) compare drug-abusing and non-drug-abusing patients on demographic and clinical variables during the acute and stabilization phases of their hospital course, and 3) obtain data from patients on their reasons for drug abuse and on acute state-related changes during periods of intoxication.

METHOD

The study group consisted of 83 psychotic inpatients consecutively admitted to the New York Hospital-Payne Whitney Clinic from May 1988 to December 1988, who gave informed consent and fulfilled the *DSM-III-R* criteria for schizophrenia ($N=68$), schizoaffective disorder ($N=12$), or schizophreniform disorder ($N=3$). The subjects were between 18 and 55 years of age, were without documented neurologic or organic brain syndrome, and were carefully screened to rule out brief drug-related psychoses. Thirteen patients with prominent histories of drug abuse were excluded when a diagnosis of schizophrenia could not be made. Two general types of patients were thereby excluded: 1) those whose exacerbation of illness occurred in the context of substance abuse and for whom psychotic symptoms remitted and did not recur during periods of abstinence from drugs and 2) patients without abstinent periods or whose functioning during such periods could not be confidently ascertained.

All diagnostic and clinical symptom assessments were performed by an independent clinical research team using the Structured Clinical Interview for *DSM-III-R* (SCID) (17), the Global Assessment Scale (GAS) (18), the Brief Psychiatric Rating Scale (BPRS) (19), and the Scale for the Assessment of Negative Sym-

toms (20). Interrater reliability (intraclass correlation coefficient) on the negative symptoms scale and the BPRS was in the 0.85–0.88 range. Patients were assessed with the BPRS, the negative symptoms scale, and the GAS during the acute phase (within 5 days of admission) and during a period of relative stability (within 5 days before discharge).

Premorbid adjustment was evaluated by using the Cannon-Spoor et al. Premorbid Adjustment Scale (21). Premorbid adjustment and demographic and family history data were gathered through interviews with the patients and, in a majority (82%) of cases, interviews with their first-degree relatives. We used a family history data form that detailed psychosis, psychiatric hospitalizations, neuroleptic treatment, serious depression, and suicide attempts of the relatives. Family histories of psychiatric disorder and drug abuse were considered to be positive if reported by any available source.

The subjects received SCID interviews to generate *DSM-III-R* diagnoses of alcohol and drug abuse and dependence (22). Information from medical records and family members provided historical data. Any patient with a *DSM-III-R* diagnosis of drug or alcohol abuse or dependence was considered to be a drug-abusing patient. Non-drug-abusing patients had no such diagnosis (current or lifetime). Data on the reliability of the research diagnoses (the drug-abusing/non-drug-abusing distinction) were based on comparison with the diagnoses made by an independent outpatient clinician (P.J.W.). Reliability was good ($\kappa=0.80$), with an estimate based on 30 cases. The clinical assessor who collected drug abuse data was blind to other clinical assessments and biological measures.

Patients who met the criteria for *DSM-III-R* drug or alcohol abuse or dependence within the 6 months before assessment were defined as recent drug abusers. Drug-abusing patients in remission were considered past drug abusers.

All patients received a semistructured drug abuse interview created for our study (23). Polysubstance-abusing patients' subjective reports of acute drug effects were generally collected for no more than two drugs. The reliability of the interview is currently being assessed for schizophrenic and borderline patients.

To assess the nonspecific characteristics of the drug-abusing group, we compared the patients with any diagnosis of drug or alcohol abuse or dependence to the patients without diagnoses of abuse or dependence. In addition, we performed three sets of post hoc analyses: 1) patients who abused only alcohol and patients who abused only cannabis were compared with patients who were not drug abusers, 2) recent drug abusers were compared with past drug abusers and patients who were not drug abusers, and 3) analyses were performed excluding the atypical schizophrenic patients (schizoaffective and schizophreniform).

We analyzed the categorical data using contingency table and chi-square analyses. Continuous data were

analyzed with t tests (two-tailed) and analysis of variance. BPRS composite scores of global symptom factors (anergia, activation, thought disorder, paranoia-suspiciousness, anxiety-depression) reported by Guy (24) were calculated. Data from the Scale for the Assessment of Negative Symptoms were analyzed using global summary items for each of the five scales. Our approach to dealing with the potential type I error associated with multiple tests was to perform a test of the overall significance using a composite measure, as recommended by Cohen and Cohen (25). Only in the case of a significant result ($p < 0.05$) for the composite did we perform tests of significance for subscales. Subjective effects data were analyzed by using a 3 (group) $\times$ 3 (response category) chi-square. We performed post hoc analyses among the groups if an overall significant ($p < 0.05$) chi-square was obtained.

RESULTS

Demographic and Substance Abuse Variables

There were no statistically significant differences between the drug-abusing and the non-drug-abusing patients on the demographic variables of age, sex, race, marital status, living arrangements, education, and occupational functioning. The patients were young (mean $\pm$ SD age = 30.6 ± 8.5 years), were unmarried (70%, $N = 58$), and had at least a high school education (78%, $N = 65$). Sixty-three percent ($N = 52$) were white. Age at onset of schizophrenia (mean $\pm$ SD = 22.9 ± 7.4 years) and number of previous hospitalizations (mean $\pm$ SD = 2.7 ± 3.3) did not differ between the two groups. The mean $\pm$ SD age at onset of alcohol use was 19.5 ± 4.7 years; age at onset of drug use was 18.9 ± 4.8 years.

Forty (48%) of the 83 patients received *DSM-III-R* diagnoses of drug abuse or dependence and met the criteria for a total of 76 diagnoses (cannabis, 26; alcohol, 21; cocaine, 14; hallucinogens, five; stimulants, five; sedative-hypnotics, three; other, two). The majority of the diagnoses were for dependence ($N = 64$) rather than abuse ($N = 12$). Nineteen patients (23%) used one substance only (cannabis, eight; alcohol, 10; stimulants, one), while 21 patients (25%) were poly-substance users. Twenty-nine (35%) of the patients were recent users. Table 1 shows the number of days within 1 week, 1 month, and 1 year before admission that recent drug abusers used alcohol, cocaine, and cannabis. Twenty-nine (48%) of the drug-abusing patients' 61 diagnoses of cannabis, alcohol, or cocaine abuse or dependence reflected daily drug use for 2 or more years. Another 12 diagnoses (20%) reflected use three times a week for 2 or more years.

Twenty-six (65%) of the 40 drug-abusing patients had family histories of substance abuse in first-degree relatives, compared with five (12%) of the 43 non-drug-abusing patients ($\chi^2 = 9.88$, $df = 1$, $p < 0.01$). In

TABLE 1. Alcohol, Cannabis, and Cocaine Use by Recent Drug-Abusing Schizophrenic Patients During the Week, Month, and Year Before Hospitalization

Period Before Hospitalization	Days of Use					
	Alcohol Abusers (N=17)		Cannabis Abusers (N=13)		Cocaine Abusers (N=7)	
	Mean	SD	Mean	SD	Mean	SD
Week	3.6	3.4	1.9	2.5	2.3	2.4
Month	14	11	11	11	9	10
Year	145	120	133	104	120	106

contrast, there were no significant differences in family history of psychiatric illness, hospitalizations, or suicide attempts between the drug-abusing and non-drug-abusing schizophrenic patients.

Clinical and Treatment Variables

There were no significant differences between the two groups of patients on any of the GAS, BPRS, or negative symptom measures at admission. However, at discharge the drug-abusing patients showed less severe psychopathology, as measured by total BPRS and negative symptom scores (table 2). Analysis of the BPRS factors revealed that drug-abusing patients had significantly lower scores on positive symptom BPRS dimensions (thought disorder and paranoia-suspiciousness) as well as anergia, but did not differ from the non-drug-abusing patients on activation and anxiety-depression.

When recent drug abusers, past drug abusers, and non-drug-abusing patients were compared, no differences were found at admission. At discharge the recent drug abusers had significantly lower mean $\pm$ SD total BPRS scores (recent abusers, 1.87 ± 0.52 ; past abusers, 1.77 ± 0.15 ; nonabusers, 2.24 ± 0.66 ; $F = 4.75$, $df = 2, 77$, $p < 0.05$) and significantly lower thought disorder scores (recent abusers, 1.78 ± 0.78 ; past abusers, 2.05 ± 1.18 ; nonabusers, 2.46 ± 1.16 ; $F = 4.99$, $df = 2, 77$, $p < 0.01$) than the non-drug-abusing patients. Both the recent and past abuser groups had significantly lower paranoia-suspiciousness scores than the nonabuser group (recent abusers, 1.64 ± 0.83 ; past abusers, 1.26 ± 0.74 ; nonabusers, 2.44 ± 1.07 ; $F = 9.81$, $df = 2, 77$, $p < 0.001$).

Analysis of scores on the Scale for the Assessment of Negative Symptoms revealed that the drug-abusing patients had significantly less affective flattening than the non-drug-abusing patients (table 2). Exclusion of the atypical schizophrenic patients resulted in the finding that besides having less affective flattening, the drug-abusing patients had less anhedonia and less avolition-apathy than the patients who were not drug abusers.

The drug-abusing and non-drug-abusing patients did not differ in global functioning as measured by the GAS, mean $\pm$ SD length of hospital stay (abusers, 47.4 ± 25.7 days; nonabusers, 61.1 ± 54.4 ; $t = -1.45$, $df = 81$, $p = 0.15$), and mean $\pm$ SD dose of neuroleptic medica-

TABLE 2. Scores on Clinical Measures at Discharge of Drug-Abusing and Non-Drug-Abusing Schizophrenic Patients

Measure	Drug Abusers (N=40)		Non-Drug-Abusers (N=43)		Significance		
	Mean	SD	Mean	SD	t	df	p
Scale for the Assessment of							
Negative Symptoms ^a							
Total	2.35	0.72	2.70	0.74	2.15	79	0.03
Alogia	2.05	0.87	2.28	1.08	1.03	79	n.s.
Anhedonia-asociality	2.74	1.06	3.12	1.18	1.52	79	n.s.
Attention	1.87	1.02	2.19	1.08	1.36	79	n.s.
Affective flattening	2.40	0.91	2.80	1.01	1.96	79	0.05
Avolition-apathy	2.71	1.20	3.12	0.91	1.76	79	0.08
BPRS ^b							
Total	1.87	0.46	2.26	0.66	2.98	78	0.004
Thought disorder	1.85	0.91	2.46	1.16	2.64	79	0.01
Paranoia-suspiciousness	1.60	0.80	2.44	1.07	3.98	79	0.001
Anxiety-depression	2.42	0.71	2.29	1.00	-0.67	79	n.s.
Anergia	1.86	0.62	2.30	0.83	2.54	78	0.02
Activation	1.65	0.61	1.73	0.82	0.49	76	n.s.
GAS ^c	43.06	8.90	40.48	12.01	-1.02	72	n.s.

^aScores range from 0 to 5; lower scores represent less severe symptoms. The total score represents the mean of the means for the five individual items.

^bScores range from 1 to 7; lower scores represent less severe symptoms. Each individual score is the mean of the means for BPRS items included in that factor; the total score represents the mean of the means for the five factors.

^cScores range from 0 (extremely dysfunctional) to 100 (extremely high level of functioning).

tion at discharge (abusers, 554.6 ± 389.7 chlorpromazine equivalents; nonabusers, 973.3 ± 1607.1 ; $t = -1.52$, $df = 73$, $p = 0.13$). (Neuroleptic doses for eight patients were unavailable because they were participants in another blind research protocol.)

Premorbid adjustment scores revealed no differences between groups in psychosocial adjustment in childhood or adulthood. However, drug-abusing patients reported significantly better sexual adjustment and worse scholastic adaptation in early adolescence (sexual adjustment: mean $\pm$ SD score for drug abusers, 1.38 ± 1.06 ; for nonabusers, 2.17 ± 1.39 ; $t = 2.16$, $df = 58$, $p < 0.05$; scholastic adaptation: drug abusers, 2.61 ± 1.42 ; nonabusers, 1.85 ± 1.31 ; $t = -2.16$, $df = 58$, $p < 0.05$) and in late adolescence (sexual adjustment: drug abusers, 2.12 ± 1.17 ; nonabusers, 2.53 ± 1.60 ; $t = 2.08$, $df = 57$, $p < 0.05$; scholastic adaptation: drug abusers, 2.74 ± 1.32 ; nonabusers, 1.97 ± 1.49 ; $t = -1.99$, $df = 54$, $p < 0.05$). (Lower scores indicate better functioning.) These differences were not apparent when only the schizophrenic patients were assessed.

Subjective Effects and Reasons for Drug Use

Cannabis, alcohol, and cocaine were the only drugs used with sufficient frequency to permit statistical analysis. Patients reported that all three drugs acutely increased happiness and decreased depression, but that alcohol, cocaine, and cannabis had different effects on anxiety, psychotic symptoms, and energy level (table 3). Alcohol and cannabis shared common effects of decreasing anxiety, in contrast to cocaine, which increased anxiety. Cannabis and cocaine were more often reported to increase energy and psychotic symptoms than alcohol. Table 4 indicates patients' reported

reasons for use, which did not differ greatly across drugs.

DISCUSSION

This study found a clinically significant lifetime prevalence of substance abuse in a sample of consecutively admitted schizophrenic inpatients: 35% of the patients had recently used drugs to an extent that caused impairment in functioning; almost 50% of the patients had experienced some impairment due to drug or alcohol use during their lifetimes. Our data are consistent with rates of drug abuse for schizophrenic patients reported by other groups in diverse geographic locations (1-3, 9). Comparison with a 16.4% lifetime prevalence rate of *DSM-III* substance abuse disorders in a community sample from the NIMH Epidemiologic Catchment Area Program (26) suggests that the rate of drug abuse among schizophrenic patients is considerably higher than that in the general population. However, interpretation of these data is inconclusive without reference to appropriately matched control groups; the prevalence of schizophrenia, as well as substance abuse, is known to be greater among younger people. Further, data from urine drug screens, if they had been available, would have strengthened the validity of our data on current use.

Our group of schizophrenic patients commonly used cannabis, alcohol, cocaine, hallucinogens, and stimulants, corroborating the findings of retrospective studies (9, 15). In comparison with a group of 19 consecutively admitted psychotic affective disorder patients of similar age in a parallel study of similar design (G. Haas, personal communication, 1990), we found that

TABLE 3. Effects of Alcohol, Cannabis, and Cocaine Use on Mood and Symptoms Reported by Drug-Abusing Schizophrenic Patients^a

Dimension	Effects of Drug									Significance	
	Alcohol Abusers (N=14)			Cannabis Abusers (N=23)			Cocaine Abusers (N=9)				
	Increased	Same	Decreased	Increased	Same	Decreased	Increased	Same	Decreased	χ ² (df=4)	p
Anxiety	0	0	14	1	0	22	6	1	2	28.64	0.001 ^b
Depression	1	3	10	1	3	19	2	3	4	5.14	n.s.
Calm	10	2	2	18	3	2	3	1	4	7.30	n.s.
Suspicion	1	11	2	17	5	1	3	3	2	17.66	0.001 ^c
Trust	3	9	1	1	11	11	1	2	5	9.46	0.05 ^d
Hallucinations	0	13	0	7	15	1	2	3	2	12.47	0.01 ^e
Slowing down	7	3	3	5	2	15	0	2	6	10.65	0.03 ^f
Stiffness	0	12	0	0	21	1	0	8	0	— ^g	—
Energy	3	0	11	16	1	5	8	0	1	15.54	0.004 ^h

^aForty-six reports on the 61 instances of dependency on or abuse of these drugs were obtained. Eight reports were missing because of lack of patient cooperation, three were missing because of temporally remote drug use, and four were missing because the patients had already reported on two other drugs.

^bCocaine versus cannabis: $\chi^2=18.7$, df=1, $p<0.0001$; cocaine versus alcohol: $\chi^2=15.7$, df=1, $p<0.001$.

^cAlcohol versus cannabis: $\chi^2=15.5$, df=1, $p<0.001$.

^dAlcohol versus cannabis: $\chi^2=7.3$, df=1, $p<0.05$.

^eAlcohol versus cannabis: $\chi^2=5.8$, df=1, $p<0.05$.

^fAlcohol versus cannabis: $\chi^2=6.7$, df=1, $p<0.05$; alcohol versus cocaine: $\chi^2=3.5$, df=1, $p=0.06$.

^gChi-square not calculable because of pattern of zeroes.

^hAlcohol versus cannabis: $\chi^2=10.9$, df=1, $p<0.01$; alcohol versus cocaine: $\chi^2=13.8$, df=1, $p<0.001$.

TABLE 4. Reasons for Drug Abuse Reported by Schizophrenic Patients With Drug-Abuse Diagnoses

Reason	Yes		No	
	N	% ^a	N	% ^a
To get high	38	72	14	26
To decrease depression	38	72	15	28
To relax	34	64	19	36
To increase pleasure	33	62	18	34
To increase energy	30	56	22	42
To go along with the group	29	55	24	45
To increase emotions	26	49	26	49
To talk more	24	45	29	55
To be more creative	21	40	31	58
To satisfy curiosity	11	21	40	75
To increase concentration	10	19	42	81
To work and study better	9	17	44	83
To decrease side effects of medication	8	15	44	83
To decrease hallucinations	6	11	46	87
To decrease suspiciousness	3	4	49	92

^aReports on 53 of the 61 instances of dependency on or abuse of these drugs were obtained. One report was missing because of lack of patient cooperation, three were missing because of temporally remote drug use, and four were missing because the patients had already reported on two other drugs. Where percentages do not add to 100, the remaining responses were "not sure."

our schizophrenic patients showed a relatively greater preference for psychotomimetic drugs (cocaine, cannabis, stimulants, and hallucinogens) in contrast to the more common use of alcohol and sedative-hypnotics among the affective disorder patients ($\chi^2=5.2$, df=1, $p<0.05$). Our findings add to the accumulating body of data indicating that schizophrenic patients preferentially abuse drugs that induce psychotic symptoms.

Our results suggest important clinical differences between drug-abusing and non-drug-abusing schizophrenic groups. The drug-abusing schizophrenic patients showed less overall psychopathology than the

schizophrenic patients without drug abuse on measures of positive and negative symptoms at discharge, although lengths of hospital stay and amounts of neuroleptic medication were equivalent. Given that the drug-abusing patients were no different from the non-abusers in psychopathology at admission, it is possible that drug-abusing schizophrenic patients respond more quickly or completely to hospital treatment than schizophrenic patients without drug abuse. The less severe symptoms demonstrated by the drug-abusing patients may also have resulted from drug abstinence in the hospital; a variety of abused drugs have been shown to exacerbate psychosis in schizophrenic patients (1, 12–14, 27, 28), and in this study recent drug-abusing patients demonstrated significantly lower levels of positive symptoms at discharge than did patients who were not drug abusers.

Even if drug abstinence did account for the lower symptom severity among the recent drug-abusing patients, the fact that equivalent treatment resulted in lower symptom severity for past drug-abusing patients requires explanation. Perhaps the social skills and motivation necessary to obtain drugs select out the more treatment-responsive schizophrenic patients with better prognosis, who, when treated appropriately and abstaining from drugs, would be less symptomatic than non-drug-abusing patients. It makes clinical sense that patients with fewer negative symptoms would be more likely to use drugs; negative symptoms might reduce the energy and motivation required to seek and obtain drugs.

When just the schizophrenic patients (excluding atypical cases) were considered, more of the negative symptoms of the drug-abusing patients were less severe at discharge. The exclusion of schizoaffective patients' data from the analysis of negative symptoms may have

reduced the possible confounding effect of depressive symptoms. Future investigations need to compare the impact of the broad and narrow definitions of schizophrenia on substance abuse patterns.

In spite of the lower severity of symptoms found in the drug-abusing patients at discharge, it is important to note that global functioning as reflected on the GAS was equivalent for the two groups. We suspect that drug abuse may impair global functioning sufficiently to compromise drug-abusing patients' possible advantage over non-drug-abusing patients in terms of less severe psychopathology.

The drug-abusing schizophrenic patients demonstrated better premorbid functioning than the patients without drug abuse on measures of sexual adjustment during early and late adolescence—a finding which confirms and extends reports of previous studies that examined adolescence (8) and premorbid *DSM-II* personality diagnoses (11). The lower scores on scholastic measures may indicate the truancy and antisocial behavior often associated with drug abuse. The superior premorbid functioning of the drug-abusing patients on social-sexual measures may indicate that schizophrenic patients with better prognostic features are more likely to have the interest and capability to seek and use drugs. It appears that the analyses of the schizophrenic patients alone did not reveal the differences found in the larger sample because of the restriction of range brought about by dropping the data of the schizoaffective disorder patients who had superior levels of premorbid functioning.

The finding that family histories of drug abuse were significantly more common in the drug-abusing than in the non-drug-abusing patients suggests that there may be familial determinants of drug abuse, which could be environmental, genetic, or both. In a study of schizophrenia and alcoholism in twins, Kendler (29) found that monozygotic twins of alcoholic schizophrenic parents were more likely than dizygotic twins of alcoholic schizophrenic parents to have both alcoholism and schizophrenia, suggesting a genetic component in both alcoholism and schizophrenia. Although our results should be interpreted cautiously because all family members were not interviewed directly and a systematic method such as the Family History Research Diagnostic Criteria procedure (30) was not used, the familial association with drug abuse in schizophrenia suggests areas for future research.

Drug-abusing patients' explanations for drug abuse suggest that schizophrenic patients may use drugs to medicate specific symptoms. Most patients reported that they used drugs to relieve depression and to relax. Responses consistent with the desire for relief of negative symptoms or depression (e.g., to increase emotions, talk more, increase energy) seemed important for about half of the patients. Relatively few patients reported abusing drugs to reduce extrapyramidal symptoms.

Drug-abusing patients' three most abused drugs (cannabis, alcohol, and cocaine) revealed distinctive

patterns of effects. The most preferred drug, cannabis, was both anxiolytic and activating (increasing energy). The next most popular drugs each shared one of those favored effects: alcohol, the anxiolytic effect, and cocaine, the activating effect. Since the drug-abusing patients reported that they abused drugs primarily to alleviate depression, perhaps they construed anxiety and decreased energy (either primary or neuroleptic-induced) as depression. The disparity between patients' reports of specific drug effects and their reasons for drug abuse may give clues about how schizophrenic patients experience depression, which is known to be difficult to distinguish from akinesia (31) and negative symptoms (32) in schizophrenia.

These self-report data are limited by several important factors. The patients' explanations of drug abuse may have been post hoc rationalizations for symptoms actually caused by drug abuse. The reliability of patients' self-report data depends on reports of their feelings and experiences while intoxicated; retrospective self-report data are subject to general biases of recall and state-dependent effects. Schizophrenic patients may not differ from other populations in this regard. Nevertheless, these perceptions, however inaccurate, may themselves drive drug-taking behavior, and thus they merit further investigation. Drug-abusing patients' reports of their reasons for drug use and the subjective effects of drugs indicate that the self-medication of dysphoria might propel some schizophrenic patients to abuse drugs.

It has been reported that drug-abusing schizophrenic patients require more hospitalization (33–35), have lower compliance with treatment regimens (3, 36, 37), and are at greater risk for suicide (38). Our data show that despite decreased scores on measures of positive and negative symptoms at discharge, drug-abusing patients function no better than patients without drug abuse during the stabilization phase, suggesting that the combination of drug abuse and schizophrenia may complicate an otherwise less severe case of schizophrenia.

REFERENCES

1. Negrete JC, Werner PK, Douglas DE, et al: Cannabis affects the severity of schizophrenic symptoms: results of a clinical survey. *Psychol Med* 1986; 16:515–520
2. Richard ML, Liskow BL, Perry PJ: Recent psychostimulant use in hospitalized schizophrenics. *J Clin Psychiatry* 1985; 46: 79–83
3. Miller FT, Tanenbaum JH: Substance abuse in schizophrenia. *Hosp Community Psychiatry* 1989; 40:847–849
4. Alterman AJ, Erdlen DL, Laporte AJ, et al: Problem drinking in hospitalized schizophrenic patients. *Addict Behav* 1980; 5:273–276
5. McLellan AT, Woody GE, O'Brien CP: Development of psychiatric illness in drug abusers: possible role of drug preference. *N Engl J Med* 1979; 301:1310–1314
6. Andreasson S, Engstrom A, Allebeck P, et al: Cannabis and schizophrenia: a longitudinal study of Swedish conscripts. *Lancet* 1987; 2:1483–1486
7. Alterman AJ, Erdlen DL, Laporte DJ, et al: Effects of illicit drug

- use in an inpatient psychiatric population. *Addict Behav* 1982; 7:231-242
8. Breakey WR, Goodell H, Lorenz PC, et al: Hallucinogenic drugs as precipitants of schizophrenia. *Psychol Med* 1974; 4:255-261
9. Weller MP, Ang PC, Latimer-Sayer DT, et al: Drug abuse and mental illness (letter). *Lancet* 1988; 1:997
10. Weller MP, Ang PC, Zachary A, et al: Substance abuse in schizophrenia (letter). *Lancet* 1984; 1:573
11. Tsuang MT, Simpson JC, Kronfol Z: Subtypes of drug abuse with psychosis. *Arch Gen Psychiatry* 1982; 39:141-147
12. Kornetsky C: Hyporesponsivity of chronic schizophrenic patients to dextroamphetamine. *Arch Gen Psychiatry* 1977; 33:1425-1428
13. Cesarc Z, Nyman AK: Differential response to amphetamine in schizophrenia. *Acta Psychiatr Scand* 1985; 71:523-528
14. van Kammen DP, Bunney WE Jr, Docherty JP, et al: *d*-Amphetamine-induced heterogeneous changes in psychotic behavior in schizophrenia. *Am J Psychiatry* 1982; 139:991-997
15. Schneider FR, Siris SG: A review of psychoactive substance use and abuse in schizophrenia: patterns of drug choice. *J Nerv Ment Dis* 1987; 175:641-652
16. Knudsen P, Vilmar T: Cannabis and neuroleptic agents in schizophrenia. *Acta Psychiatr Scand* 1984; 69:162-174
17. Spitzer RL, Williams JB, Gibbon M: Structured Clinical Interview for DSM-III-R—Patient Version (SCID-P). New York, New York State Psychiatric Institute, Biometrics Research, 1987
18. Endicott J, Spitzer RL, Fleiss JL, et al: The Global Assessment Scale: a procedure for measuring overall severity of psychiatric disturbance. *Arch Gen Psychiatry* 1976; 33:766-771
19. Overall JE, Gorham DR: The Brief Psychiatric Rating Scale. *Psychol Rep* 1962; 10:799-812
20. Andreasen NC: Negative symptoms in schizophrenia: definition and reliability. *Arch Gen Psychiatry* 1982; 39:784-788
21. Cannon-Spoor HE, Potkin SG, Wyatt RJ: Measurement of premorbid adjustment in chronic schizophrenia. *Schizophr Bull* 1982; 8:470-484
22. Spitzer RL, Williams JB: Structured Clinical Interview for DSM-III Alcohol and Drug Modules. New York, New York State Psychiatric Institute, Biometrics Research, 1987
23. Dixon L, Haas G, Weiden P, et al: Acute effects of drug abuse in schizophrenic patients: clinical observations and patients' self-reports. *Schizophr Bull* 1990; 16:69-79
24. Guy W (ed): ECDEU Assessment Manual for Psychopharmacology: Publication ADM 76-338. Washington, DC, US Department of Health, Education and Welfare, 1976
25. Cohen J, Cohen T: Multiple Regression—Correlational Analysis for the Behavioral Sciences. Hillsdale, NJ, Lawrence Erlbaum Associates, 1975
26. Regier DA, Boyd JH, Burke JD, et al: One-month prevalence of mental disorders in the United States. *Arch Gen Psychiatry* 1988; 45:977-986
27. West AP: Interaction of low-dose amphetamine use with schizophrenia in outpatients: three case reports. *Am J Psychiatry* 1974; 31:321-323
28. Treffert DU: Marijuana use in schizophrenia: a clear hazard. *Am J Psychiatry* 1978; 135:1213-1215
29. Kendler KS: A twin study of individuals with both alcoholism and schizophrenia. *Br J Psychiatry* 1985; 147:524-531
30. Endicott J, Andreasen N, Spitzer RL: Family History-Research Diagnostic Criteria. New York, New York State Psychiatric Institute, Biometrics Research, 1975
31. Van Putten T, May PR: Akinetic depression in schizophrenia. *Arch Gen Psychiatry* 1978; 35:1101-1107
32. Sommers AA: "Negative symptoms": conceptual and methodological problems. *Schizophr Bull* 1985; 11:363-379
33. Carpenter MD, Mulligan JC, Bader IA, et al: Multiple admissions to an urban psychiatric center: a comparative study. *Hosp Community Psychiatry* 1985; 36:1305-1308
34. Crowley TJ, Chesluk D, Dils S, et al: Drug and alcohol abuse among psychiatric admissions. *Arch Gen Psychiatry* 1974; 30:13-20
35. Safer DJ: Substance abuse by young adult chronic patients. *Hosp Community Psychiatry* 1987; 38:511-514
36. Kofoed L, Kahia J, Walsh T, et al: Outpatient treatment of patients with substance abuse and coexisting psychiatric disorders. *Am J Psychiatry* 1986; 143:867-872
37. Drake RE, Wallach MA: Mental patients' attitudes toward hospitalization: a neglected aspect of hospital tenure. *Am J Psychiatry* 1988; 145:29-34
38. Haas GL, Lauriello J, Dixon L, et al: Suicidal behavior in schizophrenia, in New Research Program and Abstracts, 142nd Annual Meeting of the American Psychiatric Association, Washington, DC, APA, 1989